Numerical Method - II

## Finite differences

Let $y = f(x)$ be a function, which assumes the values $f(x)$, $f(x+1)$, $f(x+2h)$, ... corresponding to the values of $x$. namely $x$, $x+h$, $x+2h$, ... the each value of $x$ is called argument and the corresponding values of $y$ is known as entry.

The first forward difference of $y = f(x)$ is defined as

$$\Delta f(x) = f(x+h) - f(x)$$

The second forward difference of $y = f(x)$ is defined as

$$\begin{aligned}\Delta^2 f(x) &= \Delta f(x+h) - \Delta f(x) \\ &= f(x+2h) - f(x+h) - [f(x+h) - f(x)] \\ &= f(x+2h) - 2f(x+h) + f(x)\end{aligned}$$

The general nth forward difference of $y = f(x)$ is defined as $\Delta^n f(x) = \Delta^{n-1} f(x+h) - \Delta^{n-1} f(x)$.

Here $\Delta$ is the forward difference operator.

If the arguments are denoted by $x_0, x_1, x_2, \ldots$ and the corresponding entries by $y_0, y_1, y_2, \ldots$ then the first, second and higher order forward differences are given by

$$\Delta y_r = y_{r+1} - y_r$$

$$\Delta^2 y_r = \Delta y_{r+1} - \Delta y_r$$

$$\Delta^3 y_r = \Delta^{n-1} y_{r+1} - \Delta^{n-1} y_r$$

The difference can be represented in the form of a table called forward difference table.

| Value of $x$ | Value of $y$ | 1st Difference | 2nd Diff | 3rd Diff | 4th Diff |
|---|---|---|---|---|---|
| $x_0$ | $y_0$ | | | | |
| | | $\Delta y_0 = y_1 - y_0$ | | | |
| $x_1 = x_0 + h$ | $y_1$ | | $\Delta^2 y_0$ | | |
| | | $\Delta y_1 = y_2 - y_1$ | | $\Delta^3 y_0$ | |
| $x_2 = x_0 + 2h$ | $y_2$ | | $\Delta^2 y_1$ | | $\Delta^4 y_0$ |
| | | $\Delta y_2 = y_3 - y_2$ | | $\Delta^3 y_1$ | |
| $x_3 = x_0 + 3h$ | $y_3$ | | $\Delta^2 y_2$ | | |
| | | $\Delta y_3 = y_4 - y_3$ | | | |
| $x_4 = x_0 + 4h$ | $y_4$ | | | | |

Here $y_0$ is the first entry is called the leading entry and $\Delta y_0$, $\Delta^2 y_0$, $\Delta^3 y_0$ and $\Delta^4 y_0$ are called the leading differences. The quantity $h$ is called the interval of differencing.

Backward differences.

Let $y_0, y_1, y_2, \ldots$ be the values of a function $y = f(x)$ corresponding to the values $x_0, x_1, x_2, \ldots$ of $x$ respectively. Let us write $\nabla y_1 = y_1 - y_0$

$$\nabla y_2 = y_2 - y_1$$

$$\nabla y_3 = y_3 - y_2$$

The $\nabla y_1, \nabla y_2, \nabla y_3 \ldots$ are called the first backward difference of $y$.

The backward difference table is given by

| Value of $x$ | Value of $y$ | 1st Diff | 2nd Diff | 3rd Diff | 4th Diff | 5th Diff |
|---|---|---|---|---|---|---|
| $x_0$ | $y_0$ | | | | | |
| | | $\nabla y_1 = \Delta y_0$ | | | | |
| $x_1 = x_0 + h$ | $y_1$ | | $\nabla^2 y_2 = \Delta^2 y_0$ | | | |
| | | $\nabla y_2 = \Delta y_1$ | | $\nabla^3 y_3 = \Delta^3 y_0$ | | |
| $x_2 = x_0 + 2h$ | $y_2$ | | $\nabla^2 y_3 = \Delta^2 y_1$ | | $\nabla^4 y_4 = \Delta^4 y_0$ | |
| | | $\nabla y_3 = \Delta y_2$ | | $\nabla^3 y_4 = \Delta^3 y_1$ | | $\nabla^5 y_5 = \Delta^5 y_0$ |
| $x_3 = x_0 + 3h$ | $y_3$ | | $\nabla^2 y_4 = \Delta^2 y_2$ | | $\nabla^4 y_5 = \Delta^4 y_1$ | |
| | | $\nabla y_4 = \Delta y_3$ | | $\nabla^3 y_5 = \Delta^3 y_2$ | | |
| $x_4 = x_0 + 4h$ | $y_4$ | | $\nabla^2 y_5 = \Delta^2 y_3$ | | | |
| | | $\nabla y_5 = \Delta y_4$ | | | | |
| $x_5 = x_0 + 5h$ | $y_5$ | | | | | |

Relation between forward and backward difference is

$$\boxed{\Delta^K y_r = \nabla^K y_{r+K}}$$

Ex:- $\Delta^2 y_3 = \nabla^2 y_5$

(1) Form the difference table and writed down $\Delta y_0, \Delta y_1, \Delta^2 y_0, \Delta^3 y_1, \nabla y_5, \nabla^2 y_4$ given that $(x_0, y_0) = (3, 12)$, $(x_1, y_1) = (7, 29)$, $(x_2, y_2) = (11, 59)$, $(x_3, y_3) = (15, 105)$ $(x_4, y_4) = (19, 165)$ $(x_5, y_5) = (23, 240)$

Soln:-

| | $x$ | $y$ | 1st Diff | 2nd Diff | 3rd Diff | 4th Diff | 5th Diff |
|---|---|---|---|---|---|---|---|
| $x_0$ | 3 | $y_0$ 12 | 17 | | | | |
| $x_1$ | 7 | $y_1$ 29 | | 13 | | | |
| $x_2$ | 11 | $y_2$ 59 | 30 | | 3 | | |
| $x_3$ | 15 | $y_3$ 105 | 46 | 16 | | -5 | |
| $x_4$ | 19 | $y_4$ 165 | 60 | | -2 | | 8 |
| $x_5$ | 23 | $y_5$ 240 | | 14 | | 3 | |
| | | | 75 | | 1 | | |
| | | | | 15 | | | |

$\Delta y_0 = 17$ , $\Delta y_1 = 30$ , $\Delta^2 y_0 = 13$ , $\Delta^3 y_1 = -2$

$\nabla y_5 = 75$ , $\nabla^2 y_4 = 14$

(2) Construct the difference table of the polynomial $x^3+5x-7$ for $x = -1, 0, 1, 2, 3, 4, 5$ and hence find $\Delta y_0$, $\Delta y_3$, $\Delta^2 y_1$ $\Delta^2 y_2$, $\nabla y_6$, $\nabla y_3$, $\nabla^2 y_5$, $\nabla^2 y_2$.

Soln:-

| | $x$ | $y$ | 1st Diff | 2nd Diff | 3rd Diff | 4th Diff |
|---|---|---|---|---|---|---|
| $x_0$ | $-1$ | $y_0$ $-13$ | | | | |
| | | | 6 | | | |
| $x_1$ | 0 | $y_1$ $-7$ | | 0 | | |
| | | | 6 | | 6 | |
| $x_2$ | 1 | $y_2$ $-1$ | | 6 | | 0 |
| | | | 12 | | 6 | |
| $x_3$ | 2 | $y_3$ 11 | | 12 | | 0 |
| | | | 24 | | 6 | |
| $x_4$ | 3 | $y_4$ 35 | | 18 | | 0 |
| | | | 42 | | 6 | |
| $x_5$ | 4 | $y_5$ 77 | | 24 | | |
| | | | 66 | | | |
| $x_6$ | 5 | $y_6$ 143 | | | | |

$\Delta y_0 = 6$, $\Delta y_3 = 24$, $\Delta^2 y_1 = 6$, $\Delta^2 y_2 = 12$, $\nabla y_6 = 66$, $\nabla y_3 = 12$

$\nabla^2 y_5 = 18$ , $\nabla^2 y_2 = 0$

Evaluate the following

(i) $\Delta \log x$ (ii) $\Delta \tan^{-1} ax$

Soln:- (i) $\Delta \log x = \log(x+h) - \log x$

$= \log\left(\frac{x+h}{x}\right)$

$= \log\left(1+\frac{h}{x}\right)$

(ii) $\Delta \tan^{-1} ax = \tan^{-1} a(x+h) - \tan^{-1} ax$ → $\tan^{-1}a - \tan^{-1}b$ formula

$= \tan^{-1}\left[\frac{ax+ah-ax}{1+a(x+h)ax}\right] = \tan^{-1}\left[\frac{ah}{1+a^2x^2+a^2xh}\right]$

# Interpolation

Interpolation is the technique of estimating the value of a function for any intermediate value of the independent variable.

Interpolation is a process of finding the value of $y$ lying between two adjacent values of $x$.

Finding the value of $f(x)$ corresponding to a value of $x$ just outside the range of $x$ is called **extrapolation**.

## Newton-Gregory forward interpolation formula:-

Let the function $y=f(x)$ take the values $y_0, y_1, y_2, \ldots$ corresponding to the values $x_0, x_0+h, x_0+2h, \ldots$ of $x$. Let it is required to find the value of $f(x)$ for $x=x_0+ph$, where $p$ is any real number

$$y = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \ldots$$

where $p = \frac{x-x_0}{h}$

Note:- the above formula is used for interpolating (and extrapolating) the values of $y$ near the beginning of the table.

## Newton-Gregory backward interpolation formula:-

$$y = y_n + p\nabla y_n + \frac{p(p+1)}{2!}\nabla^2 y_n + \frac{p(p+1)(p+2)}{3!}\nabla^3 y_n + \ldots$$

where $p = \frac{x-x_n}{h}$

Note:- The formula is normally used for interpolating the values of $y$ near the end of the table.

Reason:- Near the beginning (end) of the table sufficient number of backward (forward) differences may not be available.

⁂ Newton-Gregory forward and backward interpolation is used when the equally spaced data is given.

(1) Given

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $f(x)$ | 1 | 8 | 27 | 64 | 125 | 216 |

find $f(2.5)$.

Soln:

| $x$ | $y$ | 1st Diff | 2nd Diff | 3rd Diff | 4th Diff |
|---|---|---|---|---|---|
| $x_0$ 1 | $y_0$ 1 | $\Delta y_0$ (7) | | | |
| 2 | 8 | → | $\Delta^2 y_0$ (12) | | |
| | | 19 | → | $\Delta^3 y_0$ (6) | |
| 3 | 27 | → | 18 | → | $\Delta^4 y_0$ (0) |
| | | 37 | → | 6 | |
| 4 | 64 | → | 24 | → | 0 |
| | | 61 | → | 6 | |
| 5 | 125 | → | 30 | | |
| | | 91 | | | |
| 6 | 216 | | | | |

Now 2.5 is near the beginning of the table, hence we use Newton-Gregory forward interpolation formula.

Here $x_0 = 1$, $h = 1$, $y_0 = 1$ and $x = 2.5$

$$x = x_0 + Ph \Rightarrow P = \frac{x - x_0}{h} = \frac{2.5 - 1}{1} = 1.5$$

$$y = y_0 + P\Delta y_0 + \frac{P(P-1)}{2!}\Delta^2 y_0 + \frac{P(P-1)(P-2)}{3!}\Delta^3 y_0 + \cdots$$

$$= 1 + (1.5)(7) + \frac{(1.5)(0.5)}{2}(12) + \frac{(1.5)(0.5)(-0.5)}{6} \times 6$$

$$= 15.625$$

(2) Given that $\sin 45^\circ = 0.7071$, $\sin 50^\circ = 0.7660$, $\sin 55^\circ = 0.8192$, $\sin 60^\circ = 0.8660$. Find $\sin 52^\circ$.

Soln:- The difference table is

| | $x$ | $y$ | 1st Diff | 2nd Diff | 3rd Diff |
|---|---|---|---|---|---|
| $x_0$ | 45 | $y_0$ 0.7071 | $\Delta y_0$ 0.0589 | $\Delta^2 y_0$ −0.0057 | $\Delta^3 y_0$ −0.0007 |
| | 50 | 0.7660 | 0.0532 | −0.0064 | |
| | 55 | 0.8192 | 0.0468 | | |
| | 60 | 0.8660 | | | |

52 lies in the middle of the table, therefore you can use either backward or forward interpolation

Here $x_0 = 45$, $h = 5$, $y_0 = 0.7071$, $x = 52$

$$P = \frac{x - x_0}{h} = \frac{52 - 45}{5} = 1.4$$

By Newton-Gregory forward interpolation formula

$$y = y_0 + P\Delta y_0 + \frac{P(P-1)}{2!}\Delta^2 y_0 + \frac{P(P-1)(P-2)}{3!}\Delta^3 y_0 + \cdots$$

$$= 0.7071 + (1.4)(0.0589) + \frac{(1.4)(0.4)}{2}(-0.0057) + \frac{(1.4)(0.4)(-0.6)}{6}(-0.0007)$$

$$= 0.7880$$

(3) Find the number of students from the following data who secured marks not more than 45.

| Marks:- | 30-40 | 40-50 | 50-60 | 60-70 | 70-80 |
|---|---|---|---|---|---|
| No of Students | 35 | 48 | 70 | 40 | 22 |

Soln:- First we shall prepare cumulative frequency table

| Marks $< x$ | 40 | 50 | 60 | 70 | 80 |
|---|---|---|---|---|---|
| No of Students ($y$) | 35 | 83 | 153 | 193 | 215 |
| | | $35+48$ | $35+48+83$ | | |

The difference table is

| $x$ | $y$ | $\Delta y$ (1st Difference) | 2nd Diff | 3rd Diff | 4th Diff |
|---|---|---|---|---|---|
| $x_0$ 40 | $y_0$ 35 | (48) $\Delta y_0$ | (22) $\Delta^2 y_0$ | (−52) $\Delta^3 y_0$ | (64) $\Delta^4 y_0$ |
| 50 | 83 | 70 | −30 | 12 | |
| 60 | 153 | 40 | −18 | | |
| 70 | 193 | 22 | | | |
| 80 | 215 | | | | |

Now, $x_0 = 40$, $x = 45$, $h = 10$, $p = \frac{x - x_0}{h} = \frac{45-40}{10} = 0.5$

By Newton-Gregory forward interpolation formula

$$y = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \frac{p(p-1)(p-2)(p-3)}{4!}\Delta^4 y_0 + \cdots$$

$$= 35 + (0.5)(48) + \frac{(0.5)(-0.5)}{2}(22) + \frac{(0.5)(-0.5)(-1.5)}{6}(-52) + \frac{(0.5)(-0.5)(-1.5)(-2.5)}{24}(64)$$

$$= 35 + 24 - 2.75 - 3.25 - 2.5$$

$$= 50.5 \approx 51 \text{ students}$$

④ Evaluate $f(4.2)$ from the table given

| $x$ | 0 | 2 | 4 | 6 |
|---|---|---|---|---|
| $y$ | 2 | 10 | 66 | 218 |

Soln:- The difference table is

| $x$ | $y$ | 1st Diff $\nabla y$ | 2nd Diff $\nabla^2 y$ | 3rd Diff $\nabla^3 y$ |
|---|---|---|---|---|
| 0 | 2 | | | |
| | | 8 | | |
| 2 | 10 | | 48 | |
| | | 56 | | 48 |
| 4 | 66 | | 96 | $\nabla^3 y_n$ |
| | | 152 | $\nabla^2 y_n$ | |
| $x_n$ 6 | $y_n$ 218 | $\nabla y_n$ | | |

Now 4.2 is near the end of the table.

∴ we shall use Newton-Gregory backward interpolation formula, $x_n = 6$, $x = 4.2$, $h = 2$, $p = \frac{x - x_n}{h} = \frac{4.2 - 6}{2} = -0.9$

$$y = y_n + p\nabla y_n + \frac{p(p+1)}{2!}\nabla^2 y_n + \frac{p(p+1)(p+2)}{3!}\nabla^3 y_n$$

$$= 218 + (-0.9)(152) + \frac{(-0.9)(0.1)(96)}{2} + \frac{(-0.9)(0.1)(1.1)48}{6}$$

$$= 218 - 136.8 - 4.32 - 0.792$$

$$= 76.088.$$

⑤ The population of a town is as follows

| Year | 1921 | 1931 | 1941 | 1951 | 1961 | 1971 |
|---|---|---|---|---|---|---|
| Population in lakhs | 20 | 24 | 29 | 36 | 46 | 51 |

Estimate the increase in population during the period 1955 to 1961.

The Difference table is

| $x$ | $y$ | 1st Diff | 2nd Diff | 3rd Diff | 4th Diff | 5th Diff |
|---|---|---|---|---|---|---|
| 1921 | 20 | | | | | |
| | | 4 | | | | |
| 1931 | 24 | | 1 | | | |
| | | 5 | | 1 | | |
| 1941 | 29 | | 2 | | 0 | |
| | | 7 | | 1 | | (-9) $\nabla^5 y_n$ |
| 1951 | 36 | | 3 | | (-9) $\nabla^4 y_n$ | |
| | | 10 | | (-8) $\nabla^3 y_n$ | | |
| 1961 | 46 | | (-5) $\nabla^2 y_n$ | | | |
| | | (5) $\nabla y_n$ | | | | |
| 1971 | $y_n$ 51 | | | | | |

Here $x_n = 1971$, $x = 1955$, $h = 10$

$$P = \frac{x - x_n}{h} = \frac{1955 - 1971}{10} = -1.6$$

By Newton-Gregory backward interpolation formula

$$y = y_n + P\nabla y_n + \frac{P(P+1)}{2!}\nabla^2 y_n + \frac{P(P+1)(P+2)}{3!}\nabla^3 y_n$$

$$+ \frac{P(P+1)(P+2)(P+3)}{4!}\nabla^4 y_n + \frac{P(P+1)(P+2)(P+3)(P+4)}{5!}\nabla^5 y_n$$

$$= 51 + (-1.6)5 + \frac{(-1.6)(-0.6)}{2}(-5) + \frac{(-1.6)(-0.6)(0.4)}{6}(-8)$$

$$+ \frac{(-1.6)(-0.6)(0.4)(1.4)}{24}(-9) + \frac{(-1.6)(-0.6)(0.4)(1.4)(2.4)}{120} \times (-9)$$

$$= 51 - 8 - 2.4 - 0.512 - 0.2016 - 0.0967$$

$$= 39.7897$$

∴ Increase in population during the period 1955 to 1961 = 46 − 39.7897

= 6.2103 lakhs

(6) The following data gives the melting point of an alloy of lead and zinc, where t is the temperature in °C and P is the percentage of lead in alloy.

| P(%) | 60 | 70 | 80 | 90 |
|---|---|---|---|---|
| t | 226 | 250 | 276 | 304 |

Find the melting point of the alloy containing 84% of lead, using Newton's interpolation formula.

Soln:-

| P | t | 1st difference $\nabla$ | 2nd Diff $\nabla^2$ | 3rd Diff $\nabla^3$ |
|---|---|---|---|---|
| 60 | 226 | | | |
| | | 24 | | |
| 70 | 250 | | 2 | |
| | | 26 | | (0) $\nabla^3 y_n$ |
| 80 | 276 | | (2) $\nabla^2 y_n$ | |
| | | (28) | | |
| 90 | 304 | $\nabla y_n$ | | |

84 is near the end of the table

∴ we shall use Newton-Gregory backward interpolation

$x_n = 90$, $x = 84$, $h = 10$, $P = \frac{x - x_n}{h} = \frac{84 - 90}{10} = -0.6$

$$y = y_n + P\nabla y_n + \frac{P(P+1)}{2!}\nabla^2 y_n + \frac{P(P+1)(P+2)}{3!}\nabla^3 y_n + \cdots$$

$$= 304 + (-0.6)(28) + \frac{(-0.6)(0.4)}{2} \times 2$$

$$= 304 - 16.8 - 0.24$$

$$= 286.96$$